

Lab 9 Dbs

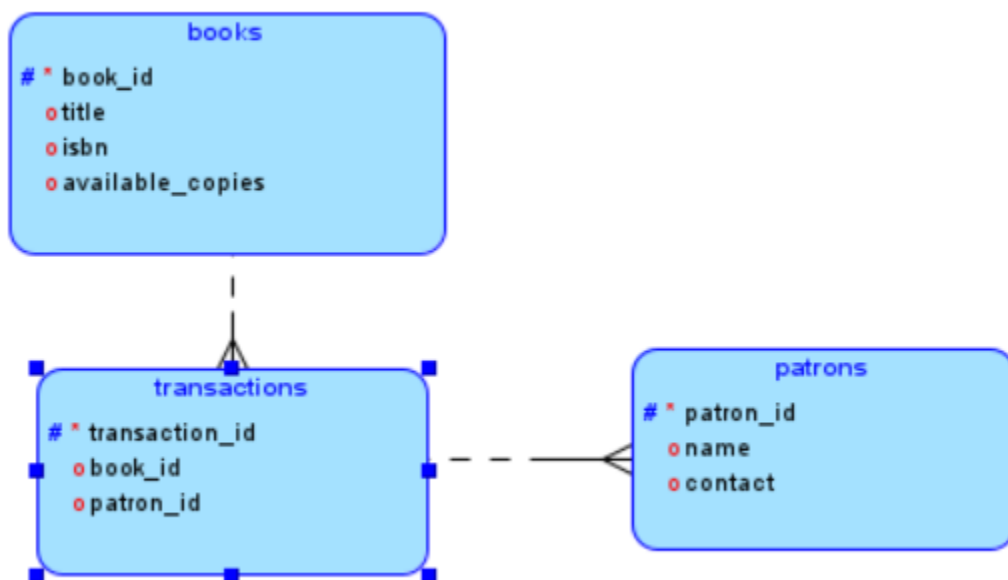
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24k-0020

Task 1:

- (a) Create a Books **entity** with attributes.
- (b) Create **Transactions entity** and define attributes.
- (c) Create relationships:

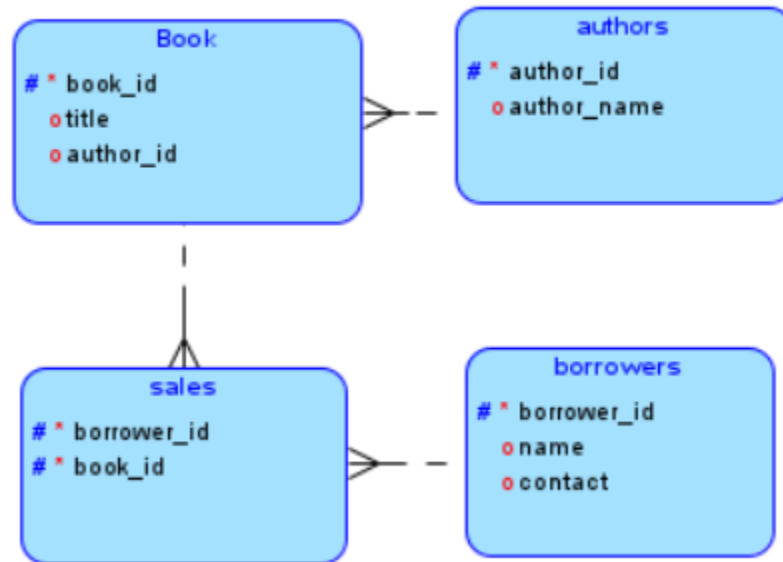
- Books → Transactions
- Patrons → Transactions



Task 2

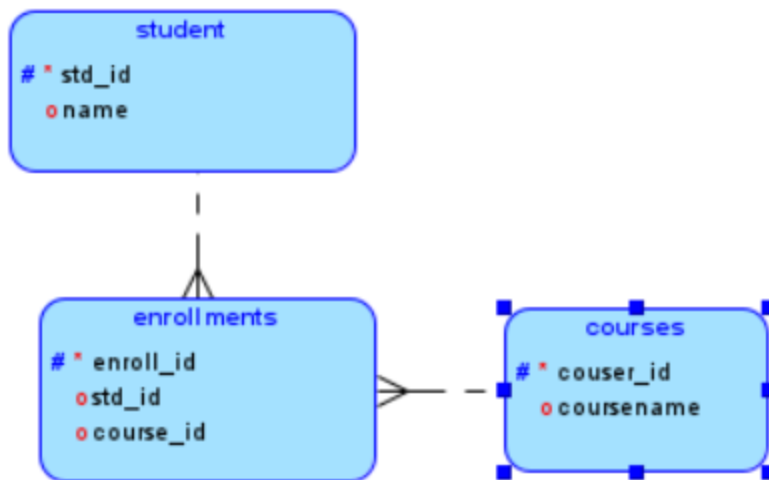
- (a) Design a **logical model** for:

- Books
- Authors
- Borrowers



(b) Create entities:

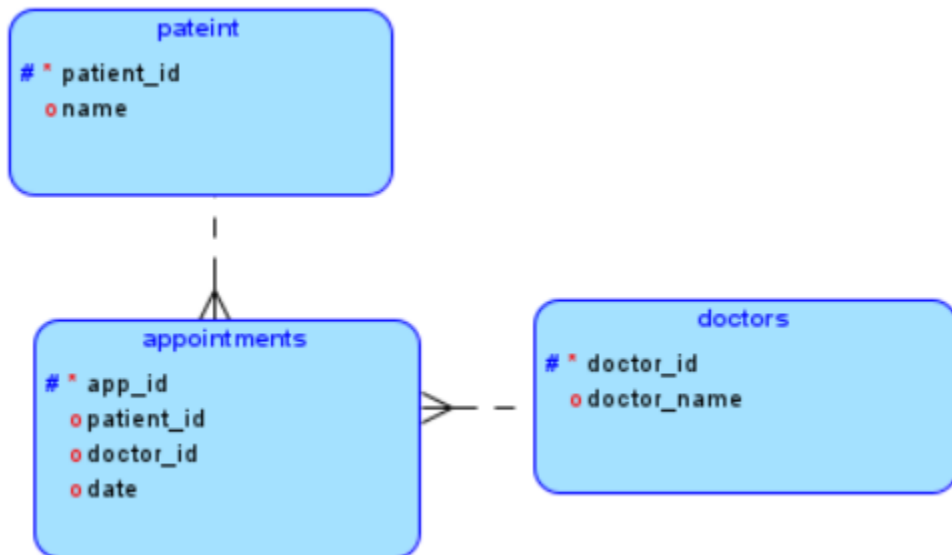
- Students
 - Courses
 - Enrollments
- Define relationships.

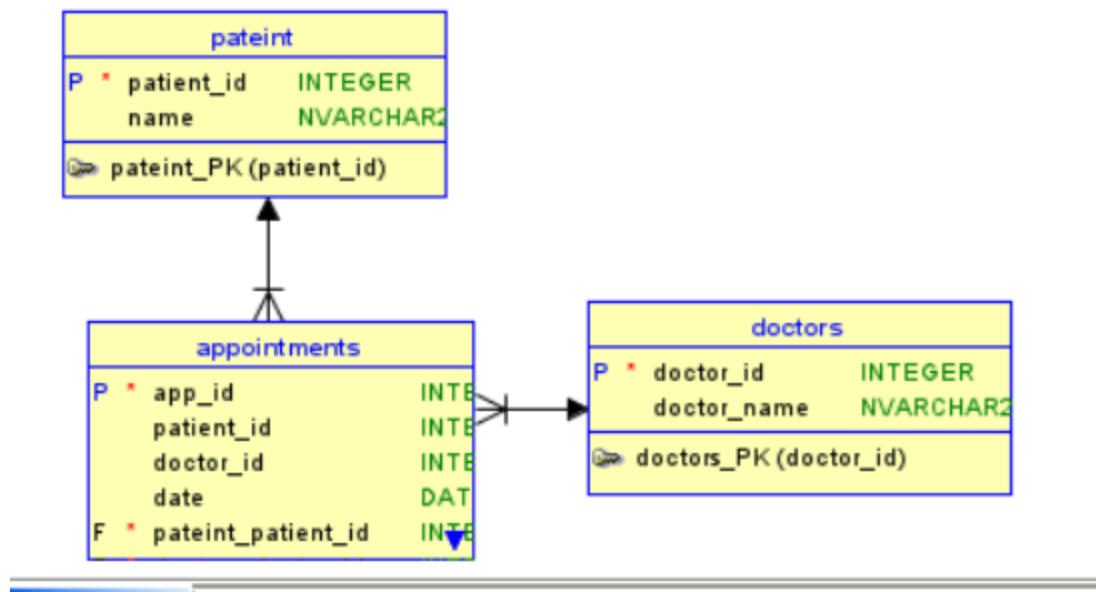


Task 3:

Design full database:

- Patients
 - Doctors
 - Appointments
- Generate **DDL script**.





DDL SCRIPT:

```
CREATE TABLE appointments
```

```
(
    app_id INTEGER NOT NULL ,
    patient_id INTEGER ,
    doctor_id INTEGER ,
    date DATE ,
    pateint_patient_id INTEGER NOT NULL ,
    doctors_doctor_id INTEGER NOT NULL
)
```

```
GO
```

```
ALTER TABLE appointments ADD CONSTRAINT appointments_PK PRIMARY KEY
CLUSTERED (app_id)
```

```
WITH (
    ALLOW_PAGE_LOCKS = ON ,
    ALLOW_ROW_LOCKS = ON )
```

```
GO
```

```
CREATE TABLE doctors
```

```
(  
    doctor_id INTEGER NOT NULL ,  
    doctor_name NVARCHAR  
)
```

```
GO
```

```
ALTER TABLE doctors ADD CONSTRAINT doctors_PK PRIMARY KEY CLUSTERED  
(doctor_id)
```

```
WITH (  
    ALLOW_PAGE_LOCKS = ON ,  
    ALLOW_ROW_LOCKS = ON )
```

```
GO
```

```
CREATE TABLE pateint
```

```
(  
    patient_id INTEGER NOT NULL ,  
    name NVARCHAR  
)
```

```
GO
```

```
ALTER TABLE pateint ADD CONSTRAINT pateint_PK PRIMARY KEY CLUSTERED (patient_id)
```

```
WITH (  
    ALLOW_PAGE_LOCKS = ON ,  
    ALLOW_ROW_LOCKS = ON )
```

```
GO
```

```
ALTER TABLE appointments
```

```
ADD CONSTRAINT appointments_doctors_FK FOREIGN KEY
```

```
(
```

```

        doctors_doctor_id
    )
    REFERENCES doctors
    (
        doctor_id
    )
    ON DELETE NO ACTION
    ON UPDATE NO ACTION
GO

ALTER TABLE appointments
    ADD CONSTRAINT appointments_pateint_FK FOREIGN KEY
    (
        pateint_patient_id
    )
    REFERENCES pateint
    (
        patient_id
    )
    ON DELETE NO ACTION
    ON UPDATE NO ACTION
GO

```

Task 4:

Design complete system:

- Logical model
- Relational model
- Physical model

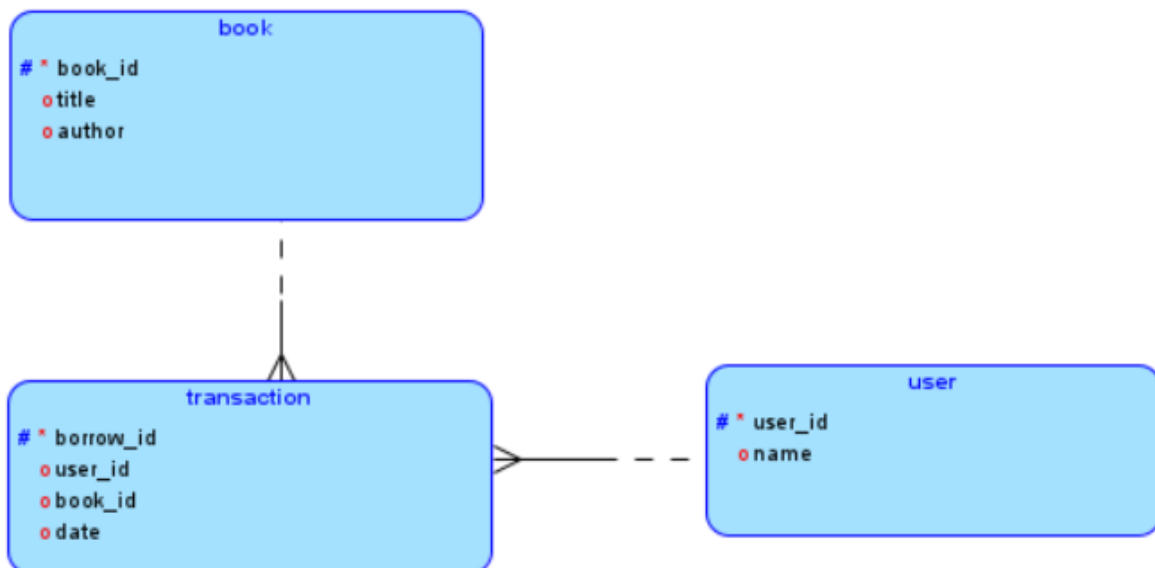
- Generate DDL

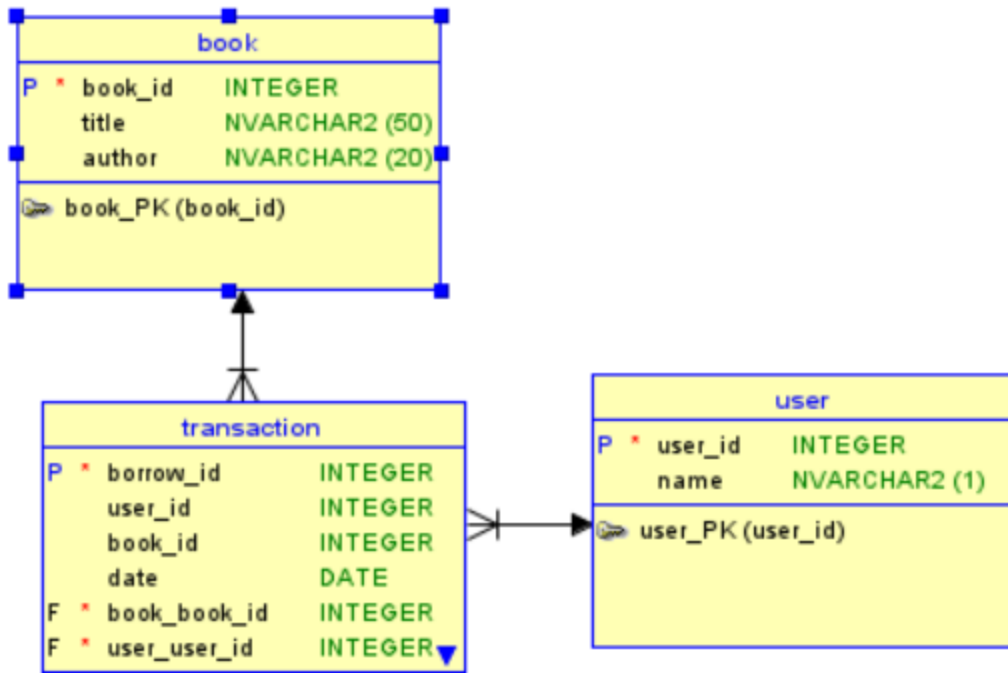
(done in task 3)

Task 5:

A software company has been assigned to design a **complete database system for a digital library platform**. The system must store information about books, users (patrons), and transactions such as borrowing and returning books.

The company wants a structured approach where the design starts from a high-level conceptual understanding and gradually transforms into an implementable database structure. The final system should be capable of generating SQL scripts that can be executed directly in a database environment.





```
CREATE TABLE book
```

```
(
```

```
  book_id INTEGER NOT NULL ,
```

```
  title NVARCHAR (50) ,
```

```
  author NVARCHAR (20)
```

```
)
```

```
GO
```

```
ALTER TABLE book ADD CONSTRAINT book_PK PRIMARY KEY CLUSTERED (book_id)
```

```
WITH (
```

```
  ALLOW_PAGE_LOCKS = ON ,
```

```
  ALLOW_ROW_LOCKS = ON )
```

```
GO
```

```
CREATE TABLE "transaction"
```

```
(
```



```
    borrow_id INTEGER NOT NULL ,
    user_id INTEGER ,
    book_id INTEGER ,
    date DATE ,
    book_book_id INTEGER NOT NULL ,
    user_user_id INTEGER NOT NULL
)
GO
```

```
ALTER TABLE "transaction" ADD CONSTRAINT transaction_PK PRIMARY KEY CLUSTERED
(borrow_id)
WITH (
    ALLOW_PAGE_LOCKS = ON ,
    ALLOW_ROW_LOCKS = ON )
GO
```

```
CREATE TABLE "user"
(
    user_id INTEGER NOT NULL ,
    name NVARCHAR
)
GO
```

```
ALTER TABLE "user" ADD CONSTRAINT user_PK PRIMARY KEY CLUSTERED (user_id)
WITH (
    ALLOW_PAGE_LOCKS = ON ,
    ALLOW_ROW_LOCKS = ON )
GO
```

```
ALTER TABLE "transaction"
ADD CONSTRAINT transaction_book_FK FOREIGN KEY
(
    book_book_id
```

```

)
REFERENCES book
(
    book_id
)
ON DELETE NO ACTION
ON UPDATE NO ACTION
GO

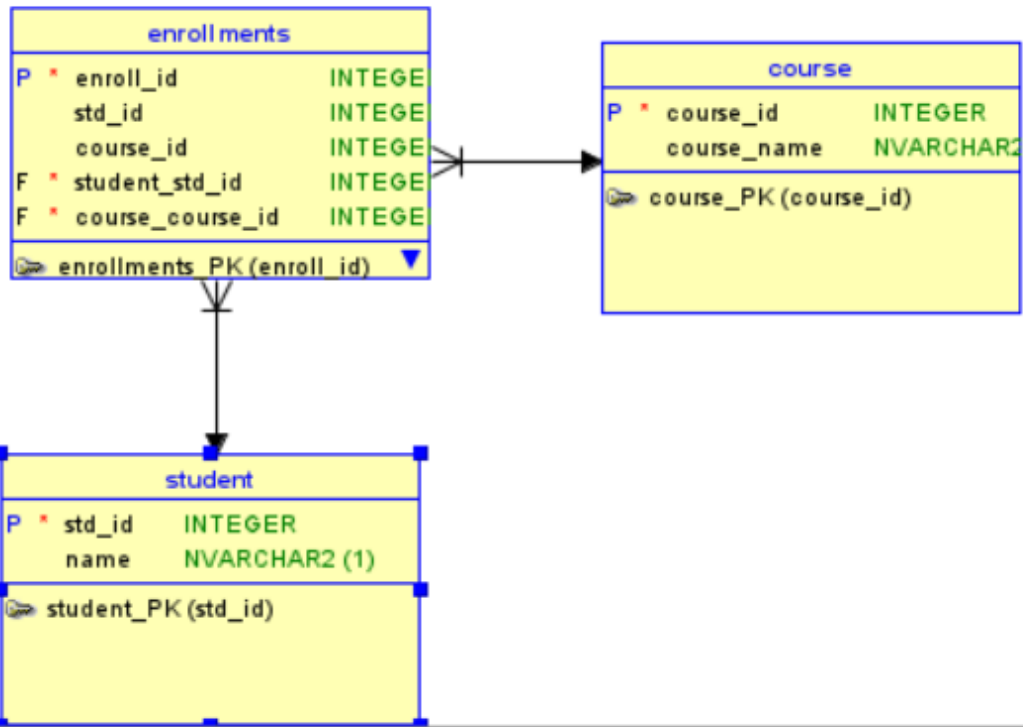
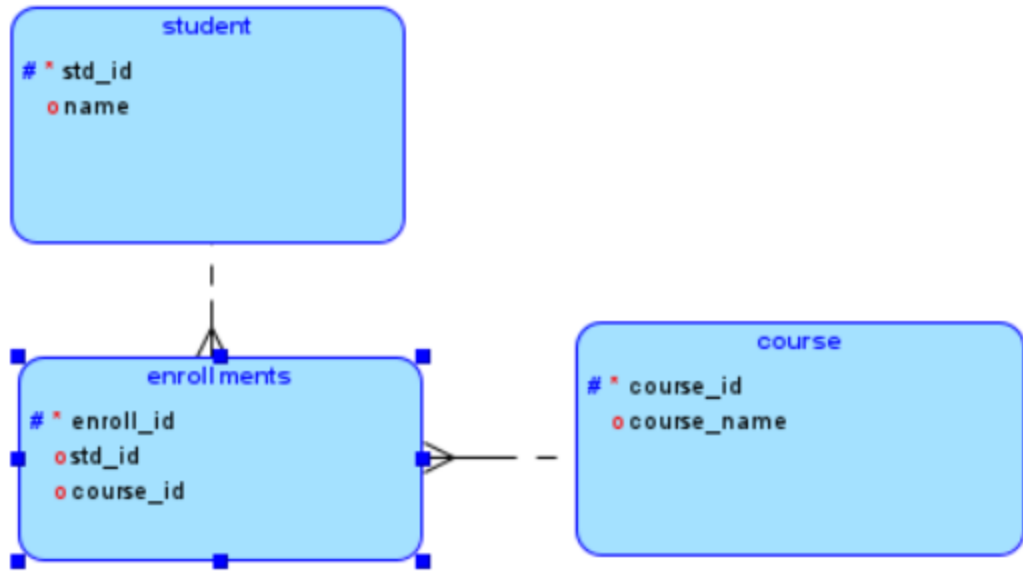
ALTER TABLE "transaction"
    ADD CONSTRAINT transaction_user_FK FOREIGN KEY
    (
        user_user_id
    )
    REFERENCES ""user""
    (
        user_id
    )
    ON DELETE NO ACTION
    ON UPDATE NO ACTION
GO

```

Task 6:

Design the complete database system by:

- Creating a logical model with entities and attributes
- Defining relationships between entities
- Transforming the logical model into a relational model
- Generating SQL (DDL) statements for implementation



```
-- at:    2026-04-25 22:21:42 PKT
-- site:   SQL Server 2012
-- type:   SQL Server 2012
```

```
CREATE TABLE course
```

```
(
    course_id INTEGER NOT NULL ,
    course_name NVARCHAR
)
```

```
GO
```

```
ALTER TABLE course ADD CONSTRAINT course_PK PRIMARY KEY CLUSTERED (course_id)
```

```
WITH (
    ALLOW_PAGE_LOCKS = ON ,
    ALLOW_ROW_LOCKS = ON )
```

```
GO
```

```
CREATE TABLE enrollments
```

```
(
    enroll_id INTEGER NOT NULL ,
    std_id INTEGER ,
    course_id INTEGER ,
    student_std_id INTEGER NOT NULL ,
    course_course_id INTEGER NOT NULL
)
```

```
GO
```

```
ALTER TABLE enrollments ADD CONSTRAINT enrollments_PK PRIMARY KEY CLUSTERED (enroll_id)
```

```
WITH (  
    ALLOW_PAGE_LOCKS = ON ,  
    ALLOW_ROW_LOCKS = ON )  
GO
```

```
CREATE TABLE student  
  
(  
    std_id INTEGER NOT NULL ,  
    name NVARCHAR  
)  
GO
```

```
ALTER TABLE student ADD CONSTRAINT student_PK PRIMARY KEY CLUSTERED (std_id)  
  
WITH (  
    ALLOW_PAGE_LOCKS = ON ,  
    ALLOW_ROW_LOCKS = ON )  
GO
```

```
ALTER TABLE enrollments  
  
    ADD CONSTRAINT enrollments_course_FK FOREIGN KEY  
  
    (  
        course_course_id  
    )  
  
    REFERENCES course  
  
    (  
        course_id  
    )  
  
    ON DELETE NO ACTION  
    ON UPDATE NO ACTION  
GO
```

```

ALTER TABLE enrollments

  ADD CONSTRAINT enrollments_student_FK FOREIGN KEY

  (
    student_std_id
  )

  REFERENCES student

  (
    std_id
  )

  ON DELETE NO ACTION

  ON UPDATE NO ACTION

GO

```

-- Oracle SQL Developer Data Modeler Summary Report:

```

--
-- CREATE TABLE          3
-- CREATE INDEX           0
-- ALTER TABLE           5
-- CREATE VIEW             0
-- ALTER VIEW              0
-- CREATE PACKAGE          0
-- CREATE PACKAGE BODY     0
-- CREATE PROCEDURE        0
-- CREATE FUNCTION         0
-- CREATE TRIGGER          0
-- ALTER TRIGGER           0
-- CREATE DATABASE         0

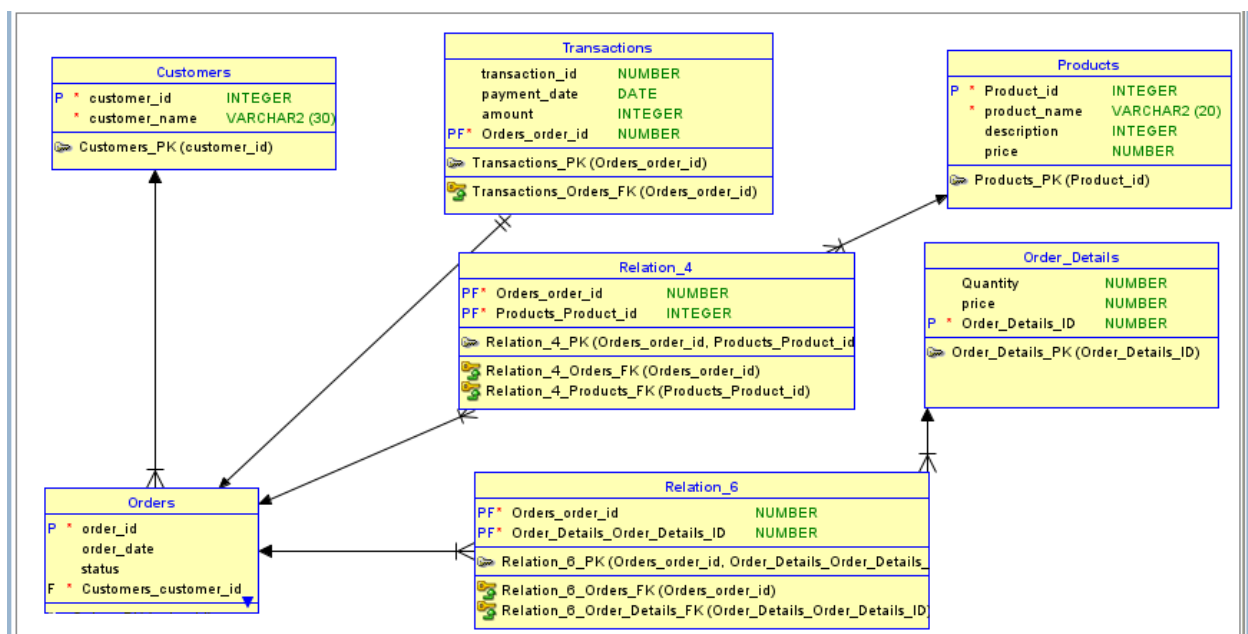
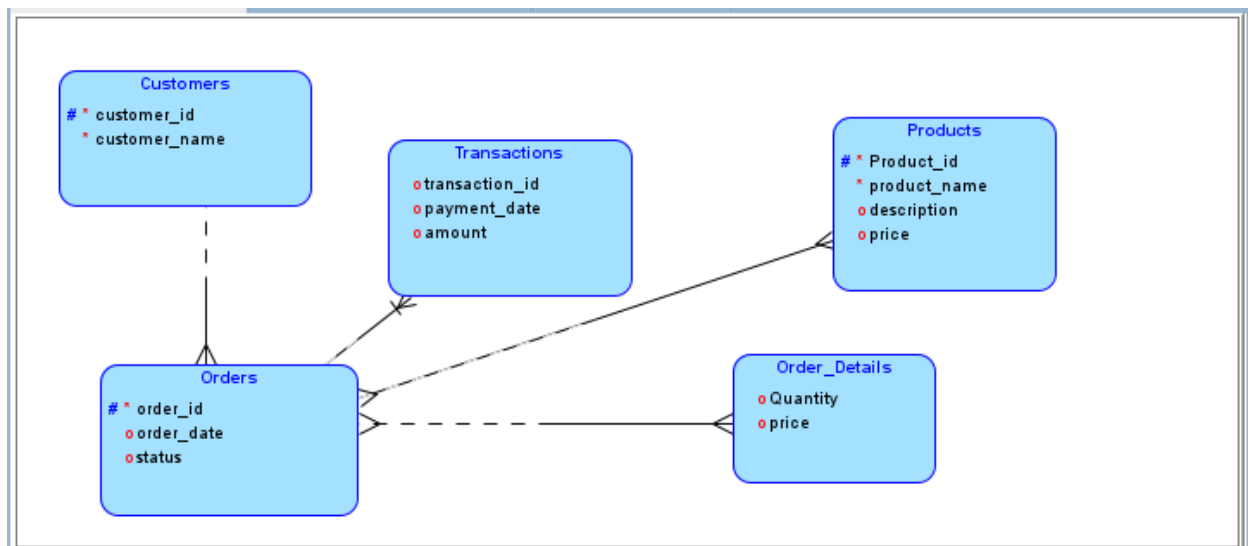
```

-- CREATE DEFAULT	0
-- CREATE INDEX ON VIEW	0
-- CREATE ROLLBACK SEGMENT	0
-- CREATE ROLE	0
-- CREATE RULE	0
-- CREATE SCHEMA	0
-- CREATE SEQUENCE	0
-- CREATE PARTITION FUNCTION	0
-- CREATE PARTITION SCHEME	0
--	
-- DROP DATABASE	0
--	
-- ERRORS	0
-- WARNINGS	0

Task 7:

A startup is building an **enterprise-level information system** that must support scalability, reusability, and consistency in data design. The system includes multiple entities such as **customers, products, orders, and transactions**, and requires standardized data definitions to ensure consistency across the database.

The design team wants to follow a professional database design pipeline, including domain creation, logical modeling, relational transformation, and physical implementation. The final output should be a deployable database schema along with a saved design for future modifications.



```
CREATE TABLE Customers
```

```
(
    customer_id  INTEGER NOT NULL ,
    customer_name VARCHAR2 (30) NOT NULL
)
```

```
;
```

```
ALTER TABLE Customers
```

```
ADD CONSTRAINT Customers_PK PRIMARY KEY ( customer_id ) ;
```



```
CREATE TABLE Order_Details
```

```
(  
    Quantity          NUMBER ,  
    price              NUMBER ,  
    Order_Details_ID  NUMBER  NOT NULL  
)
```

```
;
```

```
ALTER TABLE Order_Details
```

```
ADD CONSTRAINT Order_Details_PK PRIMARY KEY ( Order_Details_ID ) ;
```

```
CREATE TABLE Orders
```

```
(  
    order_id          NUMBER  NOT NULL ,  
    order_date        DATE ,  
    status             VARCHAR2 (10) ,  
    Customers_customer_id INTEGER  NOT NULL  
)
```

```
;
```

```
ALTER TABLE Orders
```

```
ADD CONSTRAINT Orders_PK PRIMARY KEY ( order_id ) ;
```

```
CREATE TABLE Products
```

```
(  
    Product_id  INTEGER  NOT NULL ,  
    product_name VARCHAR2 (20)  NOT NULL ,  
    description  INTEGER ,
```

```
        price          NUMBER
    )
;
```

```
ALTER TABLE Products
    ADD CONSTRAINT Products_PK PRIMARY KEY ( Product_id ) ;
```

```
CREATE TABLE Relation_4
(
    Orders_order_id      NUMBER  NOT NULL ,
    Products_Product_id  INTEGER NOT NULL
)
;
```

```
ALTER TABLE Relation_4
    ADD CONSTRAINT Relation_4_PK PRIMARY KEY ( Orders_order_id,
Products_Product_id ) ;
```

```
CREATE TABLE Relation_6
(
    Orders_order_id          NUMBER  NOT NULL ,
    Order_Details_Order_Details_ID  NUMBER  NOT NULL
)
;
```

```
ALTER TABLE Relation_6
    ADD CONSTRAINT Relation_6_PK PRIMARY KEY ( Orders_order_id,
Order_Details_Order_Details_ID ) ;
```

```
CREATE TABLE Transactions
```

```
(  
    transaction_id  NUMBER ,  
    payment_date    DATE ,  
    amount          INTEGER ,  
    Orders_order_id NUMBER NOT NULL  
)
```

```
;
```

```
ALTER TABLE Transactions
```

```
ADD CONSTRAINT Transactions_PK PRIMARY KEY ( Orders_order_id ) ;
```

```
ALTER TABLE Orders
```

```
ADD CONSTRAINT Orders_Customers_FK FOREIGN KEY
```

```
(  
    Customers_customer_id  
)
```

```
REFERENCES Customers
```

```
(  
    customer_id  
)
```

```
;
```

```
ALTER TABLE Relation_4
```

```
ADD CONSTRAINT Relation_4_Orders_FK FOREIGN KEY
```

```
(  
    Orders_order_id  
)
```

```
REFERENCES Orders
(
    order_id
)
;
```

```
ALTER TABLE Relation_4
    ADD CONSTRAINT Relation_4_Products_FK FOREIGN KEY
    (
        Products_Product_id
    )
    REFERENCES Products
    (
        Product_id
    )
;
```

```
ALTER TABLE Relation_6
    ADD CONSTRAINT Relation_6_Order_Details_FK FOREIGN KEY
    (
        Order_Details_Order_Details_ID
    )
    REFERENCES Order_Details
    (
        Order_Details_ID
    )
;
```

```
ALTER TABLE Relation_6
    ADD CONSTRAINT Relation_6_Orders_FK FOREIGN KEY
    (
        Orders_order_id
    )
    REFERENCES Orders
    (
        order_id
    )
;
```

```
ALTER TABLE Transactions
    ADD CONSTRAINT Transactions_Orders_FK FOREIGN KEY
    (
        Orders_order_id
    )
    REFERENCES Orders
    (
        order_id
    )
;
```

```
CREATE SEQUENCE Order_Details_Order_Details_ID
START WITH 1
    NOCACHE
    ORDER ;
```

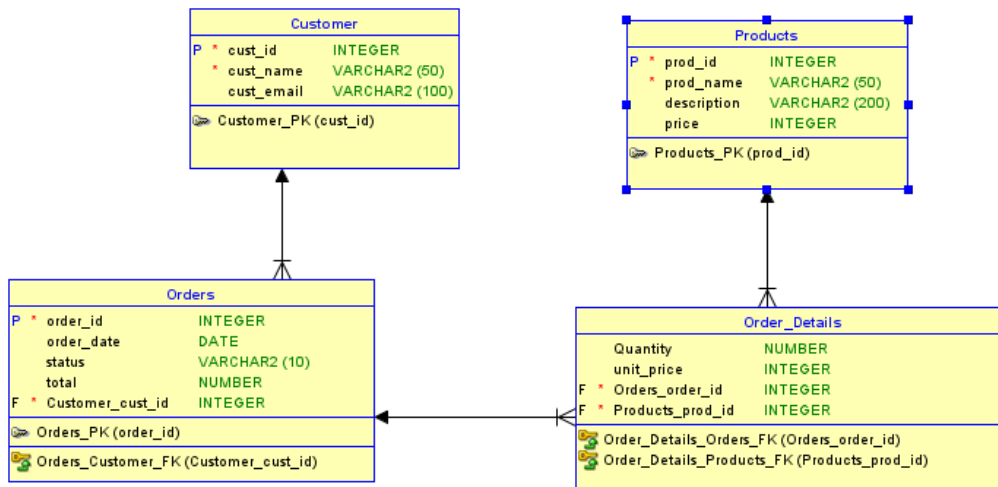
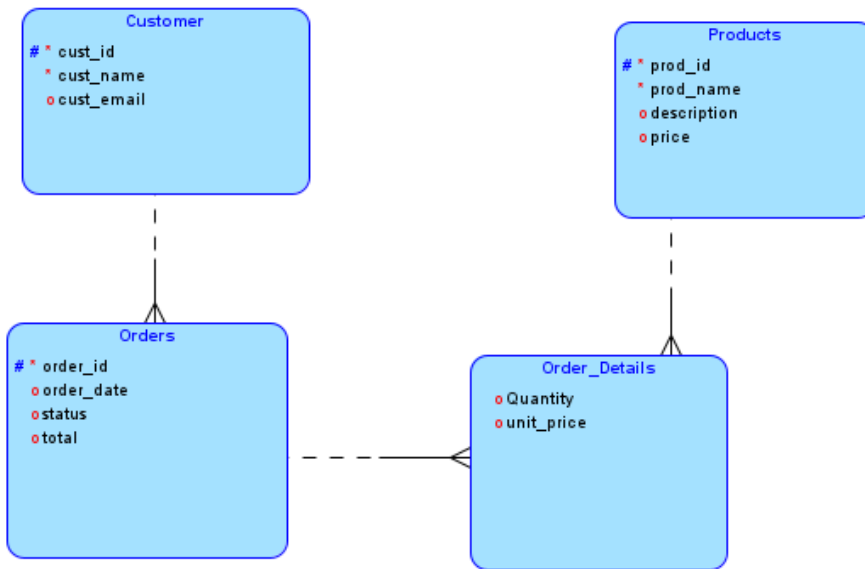
```
CREATE OR REPLACE TRIGGER Order_Details_Order_Details_ID
```

```
BEFORE INSERT ON Order_Details
FOR EACH ROW
WHEN (NEW.Order_Details_ID IS NULL)
BEGIN
    :NEW.Order_Details_ID := Order_Details_Order_Details_ID.NEXTVAL;
END;
/
```

Task 8:

Design a complete data modeling solution that:

- Defines reusable domains for common attributes
- Creates entities and relationships using a logical model
- Converts the design into a relational model
- Generates and saves DDL scripts
- Produces a final structured database design ready for deployment



Relational 1

CREATE TABLE Customer

```
(
    cust_id    INTEGER  NOT NULL ,
    cust_name  VARCHAR2 (50)  NOT NULL ,
    cust_email VARCHAR2 (100)
)
;
```

ALTER TABLE Customer

ADD CONSTRAINT Customer_PK PRIMARY KEY (cust_id) ;

CREATE TABLE Order_Details

```
(
    Quantity      NUMBER ,
    unit_price     INTEGER ,
    Orders_order_id INTEGER  NOT NULL ,
    Products_prod_id INTEGER  NOT NULL
)
;
```

CREATE TABLE Orders

```
(
    order_id      INTEGER  NOT NULL ,
    order_date     DATE ,
    status         VARCHAR2 (10) ,
    total          NUMBER ,
    Customer_cust_id INTEGER  NOT NULL
)
;
```



```
ALTER TABLE Orders
```

```
ADD CONSTRAINT Orders_PK PRIMARY KEY ( order_id ) ;
```

```
CREATE TABLE Products
```

```
(  
    prod_id      INTEGER NOT NULL ,  
    prod_name    VARCHAR2 (50) NOT NULL ,  
    description   VARCHAR2 (200) ,  
    price        INTEGER  
)
```

```
;
```

```
ALTER TABLE Products
```

```
ADD CONSTRAINT Products_PK PRIMARY KEY ( prod_id ) ;
```

```
ALTER TABLE Order_Details
```

```
ADD CONSTRAINT Order_Details_Orders_FK FOREIGN KEY
```

```
(  
    Orders_order_id  
)
```

```
REFERENCES Orders
```

```
(  
    order_id  
)
```

```
;
```

```
ALTER TABLE Order_Details
```

```
ADD CONSTRAINT Order_Details_Products_FK FOREIGN KEY
(
    Products_prod_id
)
REFERENCES Products
(
    prod_id
)
;
```

```
ALTER TABLE Orders
ADD CONSTRAINT Orders_Customer_FK FOREIGN KEY
(
    Customer_cust_id
)
REFERENCES Customer
(
    cust_id
)
;
```